

Yash Patel

 yashpatel11072025@gmail.com  +91 7046807648  Yash Patel  Portfolio

Profile

Results-driven Full Stack Software Engineer with expertise in modern JavaScript frameworks (Vue.js, Next.js, Angular) and robust backend architectures (Python, .NET Core, NestJS). Experienced in designing and architecting AI-powered search systems, offline-first Progressive Web Apps, and secure real-time communication systems. Adept at taking complex business requirements from concept to highly performant deployments.

Skills

Programming Languages

C#, JavaScript (ES6+), TypeScript, Java, Python, SQL

Frontend

Vue.js (2/3), React.js, Angular, Remix, Next.js, HTML5/CSS3, Tailwind

Backend

.NET Core Web API, Node.js, NestJS, Express.js, FastAPI

Databases & Cloud

SQL Server, PostgreSQL, MongoDB, Azure Functions

Education

07/2020 – 07/2024	B.E. (Computer Engineering)
Ahmedabad	L.D. College of Engineering
	CGPA - 8.70/10

Professional Experience

01/2024 – Present	Software Engineer TatvaSoft
Ahmedabad	Central Employee Management System: - Spearheaded the complete migration of a large-scale application from Vue 2 to Vue 3. Refactored the entire codebase to leverage the Composition API, resulting in improved maintainability and a 50% reduction in load times. Farm Harvest & Pack Management (PWA): - Architected an offline-first PWA using Vue 3, Pinia, and IndexedDB. Implemented an "outbox" pattern to queue requests during connectivity loss, ensuring 100% data integrity upon syncing. - Designed a live data synchronization module using SignalR to push real-time updates on harvest metrics and inventory changes to all active field devices. - Implemented granular Role-Based Access Control (RBAC) to securely manage user permissions and restrict access to sensitive operational data across different staff levels. Recruitment Management System: Delivered a full-stack solution (.NET Core, Angular) and implemented an Azure Function to automate email notifications and file parsing, reducing manual effort by ~80%. - Integrated frontend with backend REST APIs, establishing strict data contracts to eliminate communication errors between modules.

Projects

AI-Powered Personal Knowledge Hub [↗](#)

Next.js, NestJS, Python (FastAPI), LangChain, MongoDB Atlas Vector Search, Ollama, Socket.IO

- **Full-Stack Architecture:** Built a four-service system (Next.js web app, NestJS API, Express file storage and a stateless FastAPI + LangChain AI service) for rich-text notes with attachments, nested folders, version history with diff and restore, and Markdown/ZIP/PDF export.
- **Grounded RAG Answers:** Implemented hybrid retrieval that merges **MongoDB Atlas Vector Search** and keyword search with reciprocal rank fusion and rewrites weak queries; answers stream over **Server-Sent Events** with numbered citations that open the exact passage or PDF page.
- **Incremental Indexing Pipeline:** Built a MongoDB-backed job queue with atomic claims, retry with backoff and stale-lock recovery; content hashing re-embeds only changed passages. Reads PDFs, DOCX, and scanned pages and images through vision models.
- **Multi-Provider AI:** Shipped a built-in AI that needs no key (self-hosted Qwen models via **Ollama**) with Free/Pro token allowances and a priority queue, plus bring-your-own OpenRouter (OAuth PKCE), Gemini, OpenAI or Claude keys encrypted at rest.
- **Integrations & Real-Time:** Added a Manifest V3 browser web clipper, email-to-note and scoped API tokens; live sync across devices with **Socket.IO**; a usage analytics dashboard and a role-guarded admin panel.
- **Production Readiness:** Server-side pagination and search, upload quotas, rate limiting, ETags, health checks, graceful shutdown and structured logs with correlation IDs; an installable offline-capable PWA in English, Spanish and Hindi.

CareConnect - Hospital Management System [↗](#)

.NET Core MVC, SignalR, PostgreSQL, JWT

- **Real-Time Communication Infrastructure:** Engineered a low-latency chat module using **SignalR**, supporting 1:1 and group messaging with live "typing" status and read receipts (Sent/Delivered/Read).
- **Secure Architecture:** Implemented a granular role-based access control (RBAC) system for Admins, Physicians, and Patients using **JWT authentication** to ensure strict data privacy and HIPAA-style compliance.
- **Database Optimization:** Designed a normalized **PostgreSQL** schema to manage complex medical records, utilizing indexing strategies to reduce query execution time for patient history reports.
- **Document Handling:** Built an end-to-end file management system for securely uploading, storing, and retrieving medical reports and prescriptions.